MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – A)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions (AP)** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. The question paper consists of 13 questions divided into three Sections: A, B, and C.
3. **Section A** comprises 5 questions of 1 mark each (3 MCQs and 2 Assertion-Reason).
4. **Section B** comprises 5 questions of 2 marks each (Short Answer Type).
5. **Section C** comprises 3 questions of 5 marks each (Long Answer Type).
6. Use of calculator is not permitted.

SECTION A: Multiple Choice Questions (1 Mark Each) (5 × 1 = 5 Marks)

Q1. The common difference of the A.P. $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ is: [1]

(A) 1 (B) $\frac{1}{p}$ (C) -1 (D) $-\frac{1}{p}$

Q2. If the nth term of an A.P. is $7 - 4n$, then its common difference is: [1]

(A) 4 (B) -4 (C) 3 (D) 11

Q3. If the common difference of an A.P. is 5, then the value of $a_{18} - a_{13}$ is: [1]

(A) 10 (B) 20 (C) 25 (D) 30

Q4. **Assertion (A):** a, b, c are in A.P. if and only if $2b = a + c$.
Reason (R): The sum of the first n odd natural numbers is n^2. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

Q5. **Assertion (A):** The sum of the first fifteen terms of the A.P. $21, 18, 15, 12, \ldots$ is zero.
Reason (R): The sum of the first n terms of an A.P. is given by $S_n = \frac{n}{2}[2a + (n-1)d]$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

SECTION B: Short Answer Type Questions (2 Marks Each) (5 × 2 = 10 Marks)

Q6. Find the 9th term from the end (towards the first term) of the A.P. $5, 9, 13, \ldots, 185$. [2]

Q7. How many two-digit numbers are divisible by 3? [2]

Q8. If 7 times the 7th term of an A.P. is equal to 11 times its 11th term, then find its 18th term. [2]

Q9. Find the sum of the first 30 terms of the A.P.: $-30, -24, -18, \ldots$ [2]

Q10. If the sum of the first n terms of an A.P. is given by $S_n = 3n^2 + 5n$, find its 15th term. [2]

SECTION C: Long Answer Type Questions (5 Marks Each) ($3 \times 5 = 15$ Marks)

Q11. If the ratio of the sum of the first n terms of two A.P.s is $(7n + 1) : (4n + 27)$, find the ratio of their m^{th} terms. [5]

Q12. How many terms of the Arithmetic Progression $45, 39, 33, \ldots$ must be taken so that their sum is 180? Explain the double answer. [5]

Q13. The houses in a row are numbered consecutively from 1 to 49. Show that there exists a value of X such that the sum of the numbers of houses preceding the house numbered X is equal to the sum of the numbers of houses following it. Find this value of X. [5]

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